

PROJECT CONCEPT NOTE

Vibe Coding Masterclass Series | IBM & Bharat Cares

AWS Elastic Beanstalk: <http://studymate-ai-prod-env.eba-6yuyfhhu.ap-southeast-2.elasticbeanstalk.com>

Live HTTPS Public URL: <https://studymate-ai-5z50.onrender.com>

1. Project Title & Application Name

Application Name: StudyMate.AI

Project Title: Building & Deploying a Containerized AI Learning Platform on AWS Elastic Beanstalk & HTTPS Cloud Platform

2. Problem Statement & Objective

Students, researchers, and self-learners face significant cognitive overload when attempting to synthesize vast amounts of study notes, textbooks, and academic papers. Traditional study methods are often passive and inefficient.

The objective of StudyMate.AI is to provide an interactive, AI-driven study companion that simplifies complex lecture notes into ELI5 (Explain Like I'm 5) summaries, answers real-time student queries via an interactive tutor, and generates custom self-assessment quizzes to promote active recall.

3. Target User & Primary Use Cases

- High School & University Students: Preparing for exams, reviewing dense lecture notes, and generating custom practice quizzes.
- Researchers & Educators: Uploading research documents (.pdf, .docx, .txt) for rapid synthesis and concept breakdown.
- Lifelong Learners: Seeking simplified, structured explanations of complex subjects.

4. LLM Model & API Used

- LLM Engine: Google Gemini 3.6 Flash API
- SDK Integration: @google/genai Node.js SDK
- API Handling: Express.js backend API endpoints managing secure server-side communications to prevent client-side credential exposure.

5. Docker Containerization & AWS Architecture

- Containerization: Application containerized via multi-stage Dockerfile (Node.js 20-slim).
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6. Key Features of the Application

- **Instant Notes Simplifier & ELI5 Generator**

Transforms raw study notes or uploaded documents into structured summaries, key bullet points, and simplified explanations.

- **Multi-Format Document Ingestion**

Supports direct file uploads of PDFs, Word documents (.docx via Mammoth parser), and plain text (.txt).

- **Interactive AI Tutor Chat**

Conversational AI persona optimized for step-by-step academic explanations and guidance.

- **Automated MCQ Quiz & Score Analysis**

Generates structured multiple-choice quizzes from user notes with instant automated scoring and explanations.